Laboratory 2: Variable Speed 3 Phase Induction Motor Drives

version 1.1

Dr Mohammad Yazdani-Asrami

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Introduction

The induction motor has been the ‘workhorse of industry’ since its invention in the late 1800’s primarily due to its low maintenance and high reliability compared with early brushed motors. The one drawback of the induction machine however was its ‘fixed speed’ operation when operating off a fixed frequency AC supply. This dominance continued until the 1980’s when the development of economic power electronic converters facilitated a new generation of commercially available brushless motors (brushless DC, permanent magnet AC and switched reluctance) which offered low maintenance alternatives AND variable speed operation. Fortunately for the induction motor it too could now operate at variable speed with the inclusion of a power electronic converter thus opening up a whole new range of applications (e.g. traction) and allowing energy efficient variable speed operation in the traditional fixed speed applications (e.g. fans and pumps).

Aims and Objectives:

Model 1: Grid connected ‘fixed speed’ induction motor

- Torque vs speed curve for Grid connected operation
- Output power, input power and efficiency at a given value of slip
- Comparison with equivalent circuit calculations

Model 2: Variable speed 3 phase induction motor drive

- 3 phase inverter (DC-AC) topology
- Sinusoidal PWM control
- Variable speed control
- Constant V/Hz control and resultant torque vs speed curves

Model 1: Grid connected 'fixed speed' induction motor

The **Electrical Machines** library contains an Induction Machine model (see Figure 1) which can be applied to both motoring and generating modes of operation. We will subsequently use this model in our investigations into inverter connected variable speed operation but it would be worthwhile first having a look at it in Grid connected (fixed supply frequency) operation and validating its performance against hand calculations based on the equivalent circuit.

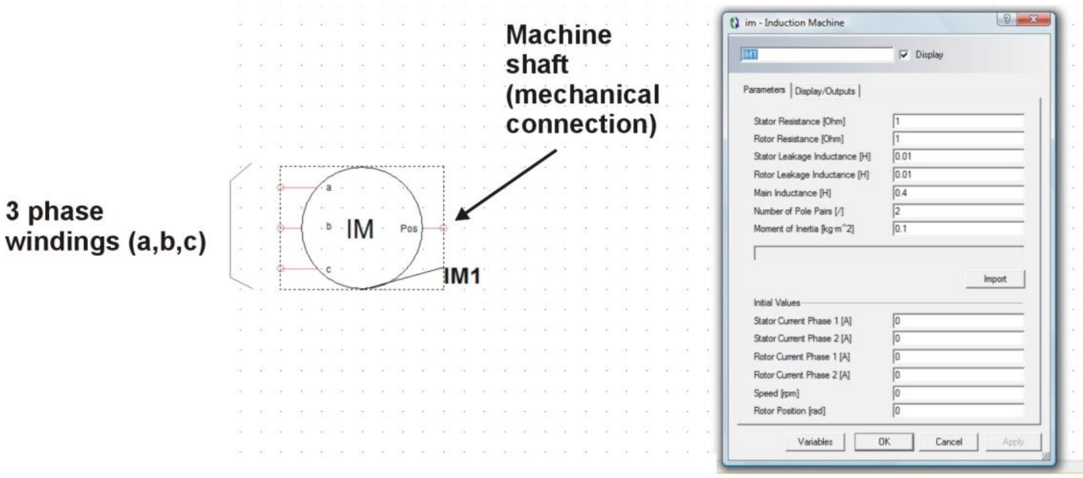


Figure 1: Portunus Induction Machine Model

There are slight differences between the Portunus equivalent circuit and the one outlined in the lecture notes so they will not give exactly the same results at a given operating point but they are close enough for the purposes for which these models are used.

Example Motor

The motor we want to investigate in this exercise has parameters outlined in Table 2 with an accompanying explanation as to how these parameters are related to equivalent circuit values (just different ways of doing the same thing!)

Portunus Model Parameter	Value		Equivalent Circuit Parameter	Value
Stator Resistance (R_s) (Ω)	0.8		Stator Resistance (R_s)	0.8Ω
Rotor Resistance (R_r) (Ω)	0.3		Rotor Resistance (R_r)	0.3Ω
Stator Leakage Inductance (L_s) (H)	2.23m		Total Leakage Reactance (X_{eq})	1.4Ω
Rotor Leakage Inductance (L_r) (H)	2.23m		Magnetising Inductance (X_m)	62.8Ω
Main Inductance (L_m) (H)	0.2		Core Loss Resistance (R_c)	ignore
Pole Pairs	3 (**NOTE**!!)		Number of Poles	6
Moment of Inertia (kg.m^2)	1			
			rms Supply Voltage (V_{srms})	240V
			Supply Frequency (f_s)	50Hz

Table 2: Induction Motor Parameters

Notes:

1. $X_{eq} = \omega.(L_s + L_r)$ (where $\omega = 2.\pi.f_s$)
2. $X_m = \omega.L_m$
3. Poles = 2.Pole_Pairs
4. The Portunus model does not include a Core Loss Resistance – I don't know why!

From this information determine the synchronous speed (N_s) for this machine operating off a 50Hz supply:

Synchronous Speed (rpm)	1000

Exercise 1.1 Torque v Speed Curve

Create a model which connects a 3-phase induction motor to a 3-phase electrical supply (E1, E2 & E3) and a mechanical speed source (NSRC1) as outlined in Figure 2 and Table 3.

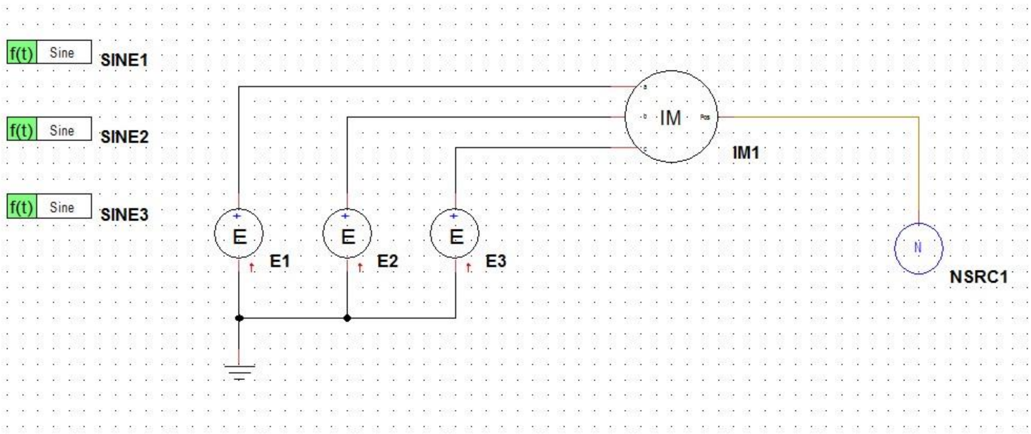


Figure 2: Grid connected induction motor

Symbol	Model	Directory	Parameters
E1	Voltage Source	Sources	Voltage –TR [V] = SINE1.OUT
E2	Voltage Source	Sources	Voltage –TR [V] = SINE2.OUT
E3	Voltage Source	Sources	Voltage –TR [V] = SINE3.OUT
SINE1	Time Function	Sine Wave	Frequency = 50 Amplitude = 339 Phase Shift = 0 Offset = 0
SINE2	Time Function	Sine Wave	Frequency = 50 Amplitude = 339 Phase Shift = 240 Offset = 0
SINE3	Time Function	Sine Wave	Frequency = 50 Amplitude = 339 Phase Shift = 120 Offset = 0
NSRC1	Speed Source	Mechanics – Rotational	Speed (rpm) = 1000

Table 3: Component Parameters

IMPORTANT Note:

1. **In Portunus the stated amplitude of the Sinewave is the PEAK value = √2. V_{rms} (where V_{rms} = 240V)**

Setup the IM1 parameters according to Table 2. Note that the phasing of the 3-phase supply (E1, E2 & E3) should look like that shown on Figure 3 – to confirm this you can put these voltages onto an On Sheet Display and simulate (TEND = 0.04) to verify.

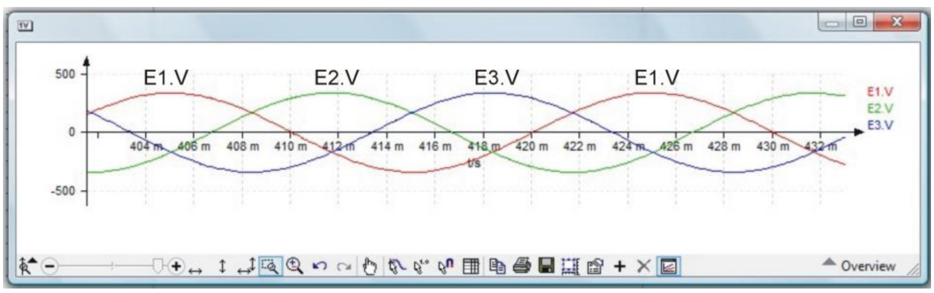


Figure 3: 3 phase grid voltages

We now want to use this model to investigate the torque v speed (slip) relationship between zero and synchronous speed. The IM1 model outputs a parameter IM1.T_SHAFT which shows instantaneous motor torque, however at the start of any simulation this value goes to a ridiculously high value (this does not happen in reality so its always worth being aware when a simulation is being silly!) so to remove this and make the output graph of torque more usable we can multiply this shaft torque with a step function which is zero for the first 0.1s of the simulation and then goes to a 1 and allows IM1.T_SHAFT to be output on MUL1 (see Figure 4). Include this in your simulation model.

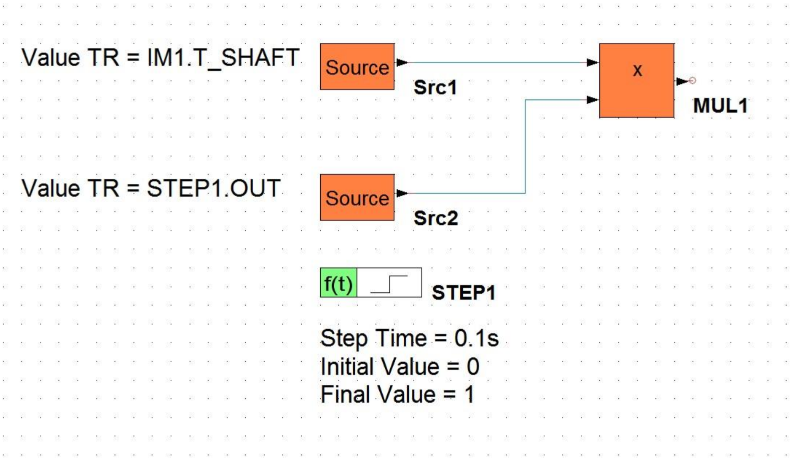


Figure 4: Removing initial spike on Shaft Torque

OK we are now ready to determine the motor torque as a function of speed. To do this we set the motor speed in the NSRC1 block, simulate at that speed and determine motor torque. Output MUL1.OUT on an On Sheet Display to display motor torque.

Set TEND = 1s in the simulation setup panel [F9] and perform a series of simulations at the following motor speeds to determine motor torque (note: as it turns out the Portunus convention is that motoring torque is negative, but for the purposes of this exercise just make it positive in the table):

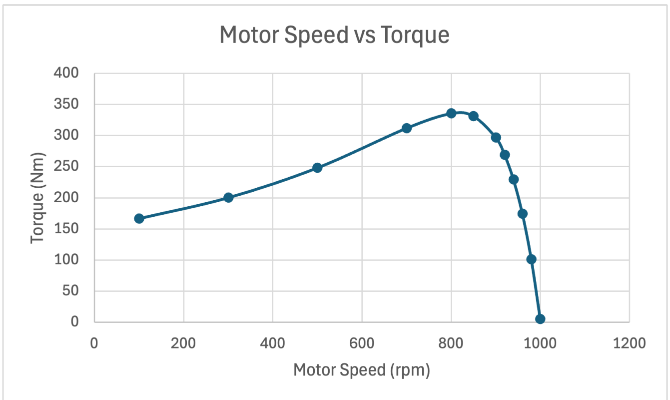
Motor Speed (rpm)	Slip	Torque (Nm)
1000	0.00	5.38
980	0.02	101.08
960	0.04	174.47
940	0.06	229.36
920	0.08	269.41
900	0.10	297.27
850	0.15	331.32
800	0.20	335.63
700	0.30	312.02
500	0.50	248.30
300	0.70	200.31
100	0.90	166.76

Note: Calculate slip for each of the motor speeds using the formula:

$$Slip(s) = \frac{N_s - N_r}{N_s}$$

Where:

N_s = Synchronous Speed
N_r = Actual motor speed



Create a graph of torque v speed from these results and determine the following:

Parameter	Value (Nm)	Speed at which this occurs (rpm)
Starting Torque (Nm)	93.60	0
Breakdown Torque (Nm)	335.63	800
Torque at Rated Slip (0.03)	136.89	970

We now want to validate the Portunus induction machine model against equivalent circuit calculations.

Exercise 1.2 Rated Operating Point Comparison

Adding additional calculation blocks (include these in Lab Report) in the Portunus model (see below for reference) determine the following for the simulation model at the 'rated operating point' for this machine:

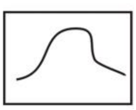
Parameter	Simulation Model	Equivalent Circuit (do this later)
Slip	0.03	0.03
Speed (rpm)	970	970
Torque (Nm)	136.89	139.20
Output (mech) Power (kW)	13.91	14.14
rms Phase Current (A)	22.45	22.04
Total input (elec) Power (kW)	15.55	15.87
Efficiency (%)	89.45	89.10

Get a demonstrator to check your results. Determine the performance based on Equivalent Circuit calculations and include the calculations and results in Lab Report.

Notes:

1. Ignore the parallel magnetising reactance and core loss resistance in the equivalent circuit calculations for simplicity.

Measuring rms values (50Hz supply):

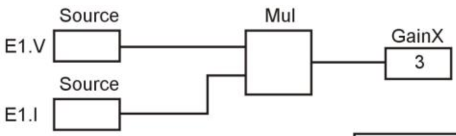


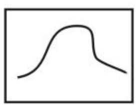
SanaX

Input Signal = set to parameter you want to measure eg E1.I
Sliding Time Window
Time = 20ms

Output SanaX.rms onto On Sheet Display

Measuring average 3 phase/50Hz electrical power (W):





SanaY

Input Signal = GainX.OUT
Sliding Time Window
Time = 20ms

Output SanaY.avg onto On Sheet Display

With this understanding of operation from a fixed frequency Grid connected induction motor we will now turn out attention to inverter connected machines.

Model 2: Inverter connected 3 phase induction motor drive

OK we now want to produce a variable speed induction motor drive using a power electronic inverter as shown on Figure 5. (note that the AC-DC rectifier shown in the lecture is not included here).

To save you time I have already created an Inverter/Motor model Lab2_Model2.bak (Fig 5) and you can access this from Moodle as outlined in the Appendix. Open this and ensure the various components are setup according to tables 4 and 5.

So why the need for all these diodes? Well 2 reasons: 1] the induction motor contains inductive elements so when switches are turned off there has to be a current path for this current to 'freewheel' as outlined in Lab 1 (D2, D4 etc), and 2] real semiconductor switches (e.g. MOSEFET, IGBT) can only conduct current in one direction so a diode (D1, D3 etc) is placed in series with each ideal switch to model this.

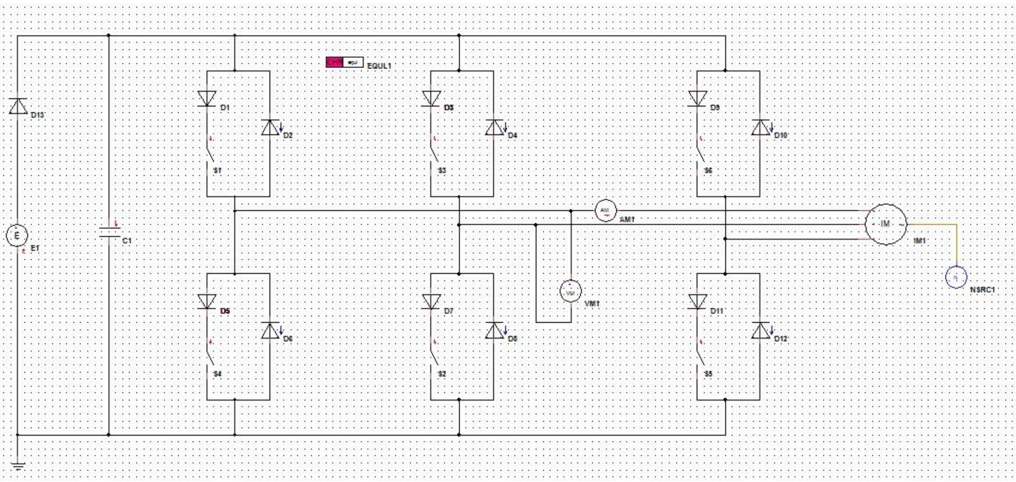


Figure 5: Variable Speed Induction Motor Drives

Set up the various model components as outlined in Tables 4 and 5.

	Model	Directory	Parameters
E1	Voltage Source	Sources	Voltage –TR [V] = 680
C1	Capacitor	Passive Components	Capacitance = 1000e-6
NSRC1	Mechanics – Rotational	Speed Source	Speed (rpm) = 970
S1-S6	Ideal Switch	Switches & Relays	Need to allocate **Control** inputs – see later
D1-D13	Diode	Semiconductors	Characteristic = EQUL1
AM1	Ammeter	Measurement Devices	
VM1	Voltmeter	Measurement Devices	

Table 4: Component Parameters

IM1 Parameter	Value
Stator Resistance (Ω)	0.8
Rotor Resistance (Ω)	0.3
Stator Leakage Inductance (H)	2.23m
Rotor Leakage Inductance (H)	2.23m
Main Inductance (H)	0.2
Number of Pole Pairs	3
Moment of Inertia (kg.m^2)	1

Table 5: Induction Motor Parameters

Sinusoidal PWM Control

The lecture outlined the principles of Sinusoidal PWM control so refer to this for a basic understanding of this control mode.
We now need to construct the Sinusoidal PWM control block in Portunus, the **per phase** implementation being shown on Figure 6.

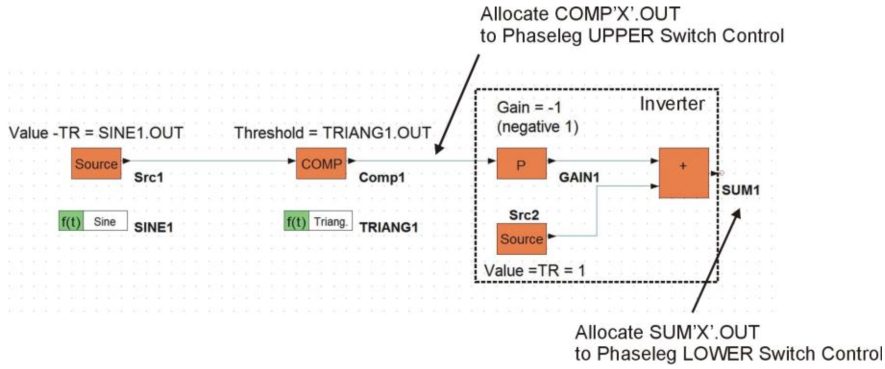


Figure 6: Per Phase Sinusoidal PWM Control

Implement these blocks for each of the 3 phaselegs according to Table 6 and allocate the relevant control signal to each of the 6 inverter switches. For each phaseleg the INVERSE of the upper switch control signal is used to control the respective lower switch. The inverter is implemented by the three blocks on the right-hand side of Figure 6. Note the -1 (negative one) gain in GAIN1 block.

Phase	Model	Directory	Parameters
1	Sine Wave (SINE1)	Time Function	Frequency (Hz) = 50 Amplitude = 1 Phase Shift = 0 Offset = 0
1	Source	Blocks	Value –TR = SINE1.OUT
1	Comparator	Blocks	Threshold = TRIANG1.OUT
2	Sine Wave (SINE2)	Time Function	Frequency (Hz) = 50 Amplitude = 1 Phase Shift = 240 Offset = 0
2	Source	Blocks	Value –TR = SINE2.OUT
2	Comparator	Blocks	Threshold = TRIANG1.OUT
3	Sine Wave (SIN3)	Time Function	Frequency (Hz) = 50 Amplitude = 1 Phase Shift = 120 Offset = 0
3	Source	Blocks	Value –TR = SINE3.OUT
3	Comparator	Blocks	Threshold = TRIANG1.OUT
All	Triangular Wave (TRIANG1)	Time Function	Frequency = 2e3 Amplitude = 1 Phase Shift = 0 Offset = 0
1	Ideal Switch (Upper)		Control Signal = Phase 1 Comp'X'.OUT
1	Ideal Switch (Lower)		Control Signal = Phase 1 Sum'A'.OUT
2	Ideal Switch (Upper)		Control Signal = Phase 2 Comp'Y'.OUT
2	Ideal Switch (Lower)		Control Signal = Phase 2 Sum'B'.OUT
3	Ideal Switch (Upper)		Control Signal = Phase 3 Comp'Z'.OUT
3	Ideal Switch (Lower)		Control Signal = Phase 3 Sum'C'.OUT

Table 6: Sine PWM Control Parameters

Exercise 2.1 Inverter based operation at Rated Operating Point

To match the conditions of the direct grid connected machine earlier we require the Inverter based drive to operate at a synchronous speed of 1000rpm and an rms phase voltage of 240V. Determine the necessary controller parameters to achieve this given the following relationships:

$$N_s = \frac{f_s \times 120}{P}$$

where:

N_s = Synch Speed (rpm)
f_s = Elec Supply Freq (Hz)
P = number of poles

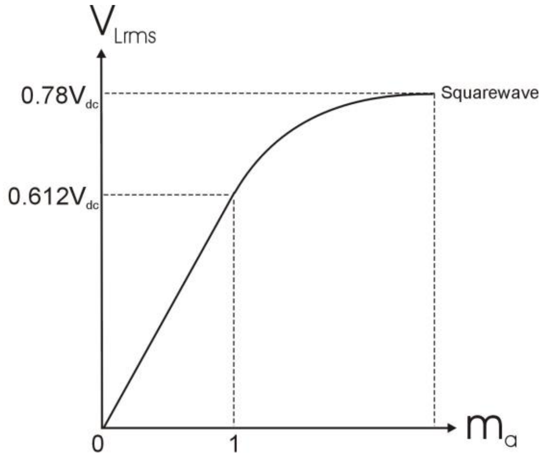


Figure 7: PWM Control

Remember:

$$Line\_Voltage(V_L) = \sqrt{3}.Phase\_Voltage(V_{ph})$$

PWM Control Parameter	Value
Sinewave Frequency (fs)	50Hz
Sinewave Amplitude (Ma)	1

Using these control values, and adding calculation and measurement blocks where necessary, determine through simulation (set TEND = 0.4s) the performance for the inverter-based drive at the rated operating point:

Parameter	Simulation Model
Slip	0.03
Speed (rpm)	970
Torque (Nm)	137.12
Output (mech) Power (kW)	13.93
rms Phase Current (A)	22.38
Average DC Power Supply Current (A)	25.39
Total input (elec) Power (kW)	17.27
Efficiency (%)	80.68%

Note: the DC Power Supply current should be averaged over a 20ms period (50Hz). Note also there is a large spike at start-up in the DC Link current so this should be removed by multiplying the current (E1.I) with a step function. (as we did to remove initial spike in motor torque output)

How these compare with the grid connected case, why might the efficiency be lower for the inverter drive case?

The inverter case has lower efficiency because the PWM switching and harmonics add extra losses, so it needs more input power to produce the same torque as the grid supply.

Exercise 2.2: Variable Speed Control

As outlined in the lecture the main purpose of connecting the inverter to an induction machine is to achieve variable speed operation. This is achieved by controlling the synchronous speed via the phase voltage frequency, but with the additional requirement of controlling the rms phase/line voltages using Ma such that a constant V/Hz ratio is maintained and saturation of the magnetic circuit avoided. What does constant V/Hz mean? Well for example if the synchronous speed is halved then the rms phase/line voltages needs to half as well. How is this achieved? Answer, by controlling the value of the Modulation Index (Ma), so for this example it has to halved too.

In this exercise we want to determine the torque v speed curves for operation around the rated operating point at 3 different synchronous speeds. What you should obtain is a set of curves which looks like the theoretical values shown on figure 8:

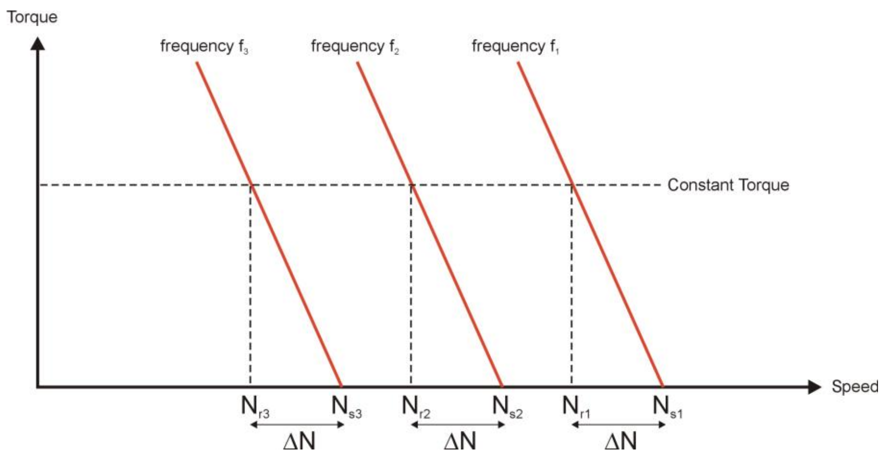


Figure 8: Torque v Speed curves for Variable Speed Operation under constant V/Hz

Constant Torque #1 = 140Nm

Replace the Constant Speed Source NSRC1 with a Constant Torque Source TRQSRC1 found in the Mechanics-Rotational Library and set the Torque –TR (Nm) parameter = -140 (NOTE THE NEGATIVE SIGN) and connect to the IM1 model – ALSO change the Triangle Oscillator frequency to 2kHz as this will quicken up the simulations.

Open the IM1 model parameters and set the **initial speed** = 1000rpm (you will subsequently have to change this to equal the synchronous speed every time you change the Synchronous speed)

Set TEND = 0.5s

Determine the necessary control parameter for each of the following 3 test points and simulate in Portunus to determine the actual motor speeds (IM1.N parameter – display this on an On Sheet Display)

Control Parameter	**Value #1**	**Value #2**	**Value #3**
Synchronous Speed (rpm)	1000	750	500
Phase Voltage Frequency fs (Hz)	50	37.5	25
Modulation Index (Ma)	1	0.75	0.5
Sinewave Frequency (Hz)	50	37.5	25
Sinewave Amplitude	1	0.75	0.5
Simulation Result:			
Actual Motor Speed (rpm)	968.8	716.6	459.0
Slip Speed ΔN (rpm)	31.21	33.4	41

Where Slip speed ΔN (rpm) = Synchronous speed (rpm) – Actual Motor Speed (rpm)

Now recognising that that torque = 0 when operating at Synchronous Speed create a graph of these results (hint – this should look something like that shown on figure 8)

Exercise 2.3: Operating Point Parameter Selection

Using theory and the graphs you have just obtained determine the required Control parameters to operate this induction motor at 800rpm **actual speed** (+/-5rpm) for a load torque of 150Nm. Simulate using these control parameters to confirm operation:

Control Parameter	**Value**
Synchronous Speed (rpm)	840
Phase Voltage Frequency fs (Hz)	42
Modulation Index (Ma)	0.84
Sinewave Frequency (Hz)	42
Sinewave Amplitude	0.84
Simulation Result:	
Actual Motor Speed (rpm)	804.6

Appendix

Accessing Existing Models

In this laboratory session you are required to access an existing Portunus model (Lab2_Model2.bak) from Moodle page ENG4187. Copy this file into a known directory on your PC. To open this file in Portunus select 'File-Open' and go to the directory where you have stored Lab2_Model2.bak. Note that in Portunus you should select 'all files' in the **Open File** panel to access the .bak file. If you subsequently want to save models then choose the relevant Lab2_Model'X'**.bak** file as it is MUCH smaller than the corresponding .ecd file but contain all the necessary information and can be subsequently opened from Portunus using the select 'all files' in the **Open File** panel.